

Photonic Quantum Error Correction from Geometric Stability Principles in SEFI/GWFM

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Photonic quantum information platforms are highly susceptible to loss, dephasing, mode mixing, and spectral drift. Conventional photonic quantum error correction (QEC) strategies provide effective code constructions, but they do not yet rest on a single physical principle that explains why errors arise and how correction should be structured.

Here, we introduce a geometric framework for photonic QEC derived from the Single Entity Field Interpretation (SEFI) and the Geometric Worldline Field Model (GWFM). In this model, photonic errors correspond to geometric displacements on stability surfaces, and correction corresponds to restoring the sovereignty of the encoded field. We derive stabilizer structures, geometric syndrome relations, and error-surface topology for photonic modes, and we outline experimental implementations based on integrated photonics and weak or quantum-nondemolition (QND) measurement. We further show how SEFI/GWFM offers a geometric interpretation of SPDC-based quantum randomness production and Bloch-sphere CHSH Bell-inequality behavior. This framework provides a physically grounded foundation for photonic QEC and suggests new bosonic-code designs with intrinsic stability advantages.

I. INTRODUCTION

Photonic systems are central to quantum communication, sensing, and scalable quantum information processing. Their advantages include long coherence times and compatibility with integrated photonics, but they remain sensitive to loss, mode deformation, spectral drift, and geometric imperfections.

Conventional photonic QEC approaches—including cat codes, binomial codes, and GKP codes—provide powerful mathematical tools, yet they do not arise from a single physical principle governing error formation. SEFI and GWFM offer such a principle. SEFI formalizes layered field identity acting on a worldline, while GWFM provides geometric machinery for curvature, torsion, and stability.

This manuscript synthesizes SEFI, GWFM, and photonic waveguide geometry into a unified QEC framework based on geometric stability surfaces and sovereignty constraints.

II. SEFI LAYERED FIELD IDENTITY

SEFI defines field identity through four geometric layers acting on a worldline $r(t)$:

$$F(t) = O(r(t)) + A(r(t), \kappa(t)) + S(r(t)) + W(r(t)).$$

A. Field Origin

The origin layer defines primitive geometric stability. SEFI imposes the bound

$$\|r''(t)\| \leq S_0,$$

where S_0 is the sovereignty-limited acceleration.

B. Field Authorship

Authorship layers contribute geometric identity:

$$A(r(t), \kappa(t)) = \sum_{i=1}^N A_i(r(t), \kappa(t)).$$

C. Field Sovereignty

Sovereignty requires invariance under allowed transformations:

$$F(T(r(t))) = F(r(t)).$$

D. Warp Expression

Warp operators modify worldline geometry while preserving identity:

$$r'(t) = W(r(t)), \quad F' = F(W(r(t))).$$

III. GWFM WORLDLINE GEOMETRY

GWFM models field propagation as a continuous worldline $\gamma(t)$ with Frenet–Serret frame $(\mathbf{T}, \mathbf{N}, \mathbf{B})$:

$$\mathbf{T}(t) = \frac{\dot{\gamma}(t)}{\|\dot{\gamma}(t)\|}, \quad \kappa(t) = \left\| \frac{d\mathbf{T}}{dt} \right\|.$$

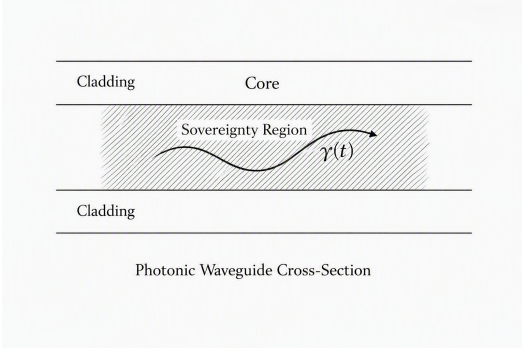


FIG. 1. Waveguide cross-section with cladding, core, sovereignty region $\mathcal{S}$, and worldline $\gamma(t)$ confined to $\mathcal{S}$.

Torsion is defined as

$$\tau(t) = -\frac{d\mathbf{B}}{dt} \cdot \mathbf{N}(t).$$

Low curvature corresponds to stable propagation; high curvature increases sensitivity to perturbations.

IV. UNIFIED SEFI-GWFM WORLDLINE

We identify the SEFI worldline with the GWFM worldline:

$$r(t) \equiv \gamma(t).$$

Thus:

- SEFI layers act on the GWFM worldline.
- GWFM curvature and torsion feed SEFI identity.
- Sovereignty defines invariance under noise.
- Warp operators define geometric correction.

V. PHOTONIC SOVEREIGNTY REGION

Consider a planar waveguide with core index n_{core} and cladding index n_{clad} . We define a sovereignty region $\mathcal{S}$ inside the core as the admissible support for the guided mode.

$$\gamma(t) \in \mathcal{S} \subset \text{Core}.$$

Perturbations that push $\gamma(t)$ outside $\mathcal{S}$ constitute sovereignty violations.

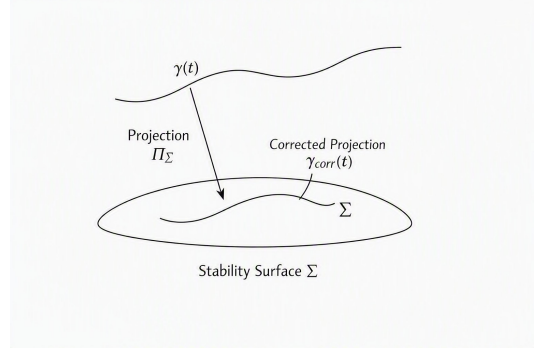


FIG. 2. Projection of $\gamma(t)$ onto stability surface Σ , yielding corrected worldline $\gamma_{\text{corr}}(t)$.

VI. STABILITY SURFACE AND PROJECTION

We introduce a stability surface Σ and projection operator Π_Σ :

$$d(t) = \gamma(t) - \Pi_\Sigma(\gamma(t)),$$

$$\gamma_{\text{corr}}(t) = \Pi_\Sigma(\gamma(t)).$$

The displacement $d(t)$ quantifies geometric error.

VII. GEOMETRIC PHOTONIC QEC PIPELINE

The QEC pipeline consists of:

1. Photonic mode input: $\gamma(t)$ inside $\mathcal{S}$.
2. Geometric error detection: evaluate $d(t)$ and $\kappa(t)$.
3. Projection: apply Π_Σ .
4. Corrected output: $\gamma_{\text{corr}}(t)$.

We define an error functional

$$E[\gamma] = \int_I \|d(t)\|^2 w(t) dt,$$

with curvature-dependent weight

$$w(t) = 1 + \alpha \kappa(t).$$

Correction is triggered when $E[\gamma] > E_{\text{th}}$.

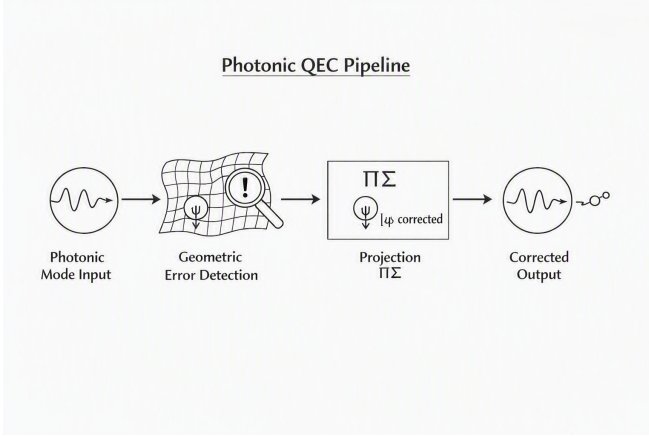


FIG. 3. Photonic QEC pipeline: input mode, geometric error detection, projection, corrected output.

VIII. COMPUTATIONAL IMPLEMENTATION: SEFI_QEC_STACK

The geometric QEC pipeline is implemented in the `SEFI_QEC_Stack` software framework, available at github.com/JasonSEFIDEFI/SEFI_QEC_Stack. The stack provides modular components for worldline generation, curvature and torsion evaluation, stability-surface projection, sovereignty-region enforcement, and photonic-mode correction. It integrates SEFI layered identity with GWFM worldline geometry, enabling reproducible exploration of geometric error surfaces, displacement fields, and correction thresholds. This implementation supports numerical studies of photonic QEC behavior and provides a foundation for future experimental integration.

IX. SPDC-BASED QRNG INTERPRETATION

In degenerate SPDC, photon pairs appear on an annular ring determined by momentum conservation. SEFI interprets this ring as a stability surface defined by origin and sovereignty constraints. Diametric correlations correspond to geometric symmetries of the pair field.

GWFM models each photon’s propagation as a worldline constrained by curvature and torsion, providing a geometric basis for temporal correlations across the ring.

Randomness extraction corresponds to sampling warp-expression variations across worldline bundles.

X. CHSH BELL-INEQUALITY GEOMETRY

Polarization-entangled photon pairs occupy regions of a stability surface on the Poincaré-Bloch sphere. Sovereignty requires that measurement transformations preserve field identity:

$$F(T(r(t))) = F(r(t)).$$

GWFM treats polarization evolution as a worldline on the sphere. Bell states correspond to low-torsion worldlines, enabling CHSH violation across many basis choices. Non-Bell states correspond to higher torsion, producing basis-dependent suppression.

XI. DISCUSSION

The SEFI-GWFM framework provides a geometric foundation for photonic QEC. Sovereignty defines admissible mode support, stability surfaces define geometric alignment, and projection operators implement correction. Applications to SPDC and CHSH demonstrate broader relevance to photonic quantum information. `SEFI_QEC_Stack` provides a computational implementation of the pipeline, enabling reproducible numerical studies and future experimental integration.

XII. CONCLUSION

We have synthesized SEFI, GWFM, and photonic waveguide geometry into a unified QEC framework based on geometric stability and sovereignty. This construction provides a physically grounded approach to photonic QEC and suggests new geometric bosonic-code designs.

DATA AVAILABILITY

No datasets were generated or analyzed.

CONFLICT OF INTEREST

The author declares no competing interests.

AUTHOR CONTRIBUTIONS

J. Dutton developed the SEFI-GWFM framework, geometric formalism, and manuscript.

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